

Proof. Let

$$f(t) = \sum_{i=1}^n (a_i - t b_i)^2, \quad t \in \mathbb{R}.$$

Each summand is a square, hence $f(t) \geq 0$ for all t .

Expanding $f(t)$ using $(a - tb)^2 = a^2 - 2tab + t^2b^2$ gives

$$\begin{aligned} f(t) &= \sum_{i=1}^n (a_i^2 - 2ta_i b_i + t^2 b_i^2) \\ &= \left(\sum_{i=1}^n b_i^2 \right) t^2 - 2 \left(\sum_{i=1}^n a_i b_i \right) t + \sum_{i=1}^n a_i^2. \end{aligned}$$

Set

$$A := \sum_{i=1}^n b_i^2, \quad B := -2 \sum_{i=1}^n a_i b_i, \quad C := \sum_{i=1}^n a_i^2,$$

so that $f(t) = At^2 + Bt + C$.

Since $f(t) \geq 0$ for every real t , the quadratic polynomial must have non-positive discriminant:

$$B^2 - 4AC \leq 0. \tag{1}$$

Substituting the expressions for A, B, C into (1) yields

$$\left(-2 \sum_{i=1}^n a_i b_i \right)^2 - 4 \left(\sum_{i=1}^n b_i^2 \right) \left(\sum_{i=1}^n a_i^2 \right) \leq 0.$$

Dividing by $4 > 0$ and simplifying gives

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right),$$

which is the Cauchy–Schwarz inequality.

Equality condition. Equality in (1) occurs iff the discriminant is zero, i.e. the quadratic has a double root. In that case

$$f(t) = A(t - t_0)^2, \quad t_0 = \frac{\sum_{i=1}^n a_i b_i}{\sum_{i=1}^n b_i^2}.$$

Because $f(t) = \sum_{i=1}^n (a_i - tb_i)^2$, the equality $f(t_0) = 0$ forces each term to vanish:

$$a_i = t_0 b_i \quad \text{for all } i.$$

Conversely, if there exists $\lambda \in \mathbb{R}$ with $a_i = \lambda b_i$ for every i , then every summand $(a_i - \lambda b_i)^2$ is zero, so $f(\lambda) = 0$ and the discriminant is zero. Hence equality holds precisely when the vectors $(a_1, \dots, a_n)$ and $(b_1, \dots, b_n)$ are linearly dependent. $\square$